

CoreSys Executable System Revision Report

1. Executable File Format Revision

The CoreSys executable format will use a simplified design. Instead of a custom executable container with headers and metadata, .bin files will contain only raw machine code.

The file structure is:

program.bin

- Machine Code Instructions only

The kernel will not require:

- Magic numbers
- Version fields
- Entry metadata
- Size fields
- Data sections

The first byte of the file represents the first instruction executed.

2. Binary Loading Process

Execution flow:

Shell

- > Kernel
- > Load program.bin
- > Allocate memory
- > Copy binary
- > Jump to start address
- > Program execution
- > Return to kernel/shell

3. Program Return Model

CoreSys programs always return control to the kernel or shell after execution.

Lifecycle:

1. Kernel loads program.
2. Kernel transfers execution.
3. Program runs.
4. Program returns.
5. Kernel cleans resources.
6. Shell continues.

4. Task Management Revision

The initial task model does not require complex process scheduling.

CoreSys uses a simple execution model:

Kernel Task

- > Load Program
- > Execute
- > Return to Kernel

The kernel remains responsible for execution control.

5. Advantages

Simpler loader:

- Less code
- Easier debugging
- Faster development
- Lower complexity

Direct hardware-level execution:

- Compiler can target CoreSys directly.
- Binary format is predictable.

6. Technical Considerations

The kernel must manage:

- Loading address
- Execution start point
- Program return mechanism

Possible approaches:

- Fixed load address
- Dedicated return-to-kernel instruction

7. Updated Architecture

Shell

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CoreSys Kernel

- Binary Loader
- Memory Manager
- Return Handler

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program.bin

- Raw Machine Code

8. Overall Evaluation

The simplified executable design reduces complexity and is suitable for early operating system development. The kernel takes more responsibility for loading, memory handling, compatibility, and safe returns.

Conclusion:

CoreSys .bin files are raw executable binaries containing only machine code. Programs execute and then return control to the kernel or shell, creating a simple and controlled execution model.